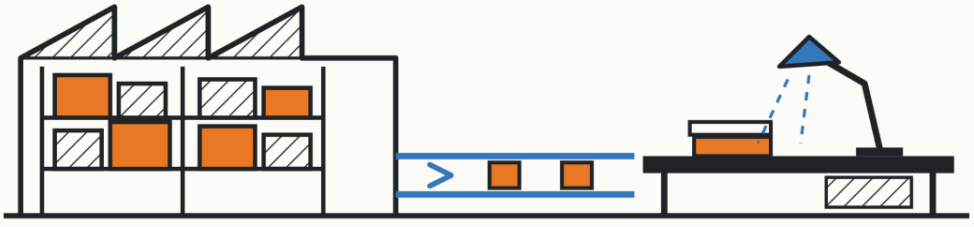
Transfers alone miss the target

Produce all, then reduce · analytical
Assumed 8 decimal TB/s, shared



Target 0.200 ms

Transfer floor ≥0.243 ms



1,856 MiB ÷ assumed 8 TB/s.
Transfers alone exceed this target.
Compute and other traffic excluded.

Scratch becomes traffic

Attention backward · one sequence, one layer · one device · FP32 partials



1,056 MiB produced

≤128 MiB retained on chip

≥1,856 MiB out + back



All production, then reduction.
≥928 MiB out + ≥928 MiB reread.
No compression or recomputation.

Compare the complete path



Store all partials

Simple ownership.

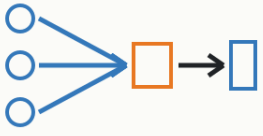
More scratch + traffic.



Combine smaller batches

Less live scratch.

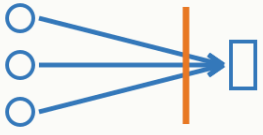
More coordination.



Accumulate directly

Less staging.

Check synchronization.



Each design needs correctness checks.

Sum order can change rounding.